

ASTRA Mission Concept

T-REX: Time-Resolving Explorer

Authors: Ref Bari (Brown University, refath_bari@brown.edu), Chompoonek (Chicha) Nimitpornsuko (Brown University, chompoonek_nimitpornsuko@brown.edu), Sharanya Palit (Brown University, sharanya_palit@brown.edu), Henry Tan (Brown University, henry_tan@brown.edu), Neil Stringer (Brown University, neil_stringer@brown.edu), Amelia Frickey (Brown University, amelia_frickey@brown.edu), Graham Neely (Brown University, graham_neely@brown.edu)

Advisors: Lucas Anderson (Utah State University, lucas.anderson@usu.edu), Locke Spencer (locke.spencer@uleth.ca), Farhad Zadeh (zadeh@northwestern.edu), Kimberly Weaver (Goddard, kimberly.a.weaver@nasa.gov), Paul Goldsmith (paul.f.goldsmith@jpl.nasa.gov), Carlos Saavedra (saavedra@queensu.ca)

For CoPAG & PhysPAG



T-REX: Time-Resolving Explorer

1. Science Investigation

The Astro2020 Decadal Survey identifies NASA's time-domain and multi-messenger program as the agency's "highest-priority sustaining activity". To this end: In 2015, LIGO made the first direct detection of gravitational waves, followed shortly by the EHT's direct image of a black hole in 2019. The upcoming decade welcomes precision tests of general relativity with unprecedented accuracy, enabled by sub-mm imaging of supermassive black holes (by ngEHT) and optical-infrared imaging of Schwarzschild precession in the galactic center (by GRAVITY+). In parallel, gravitational wave detectors spanning the entire nHz-mHz frequency band will be coming online: NANOGrav is building compelling evidence for a population of "local SMBHs" contributing to the stochastic GW background, whilst LISA will search for Extreme-Mass Ratio Inspirals (EMRIs), enabling tests of beyond-Kerr metrics.

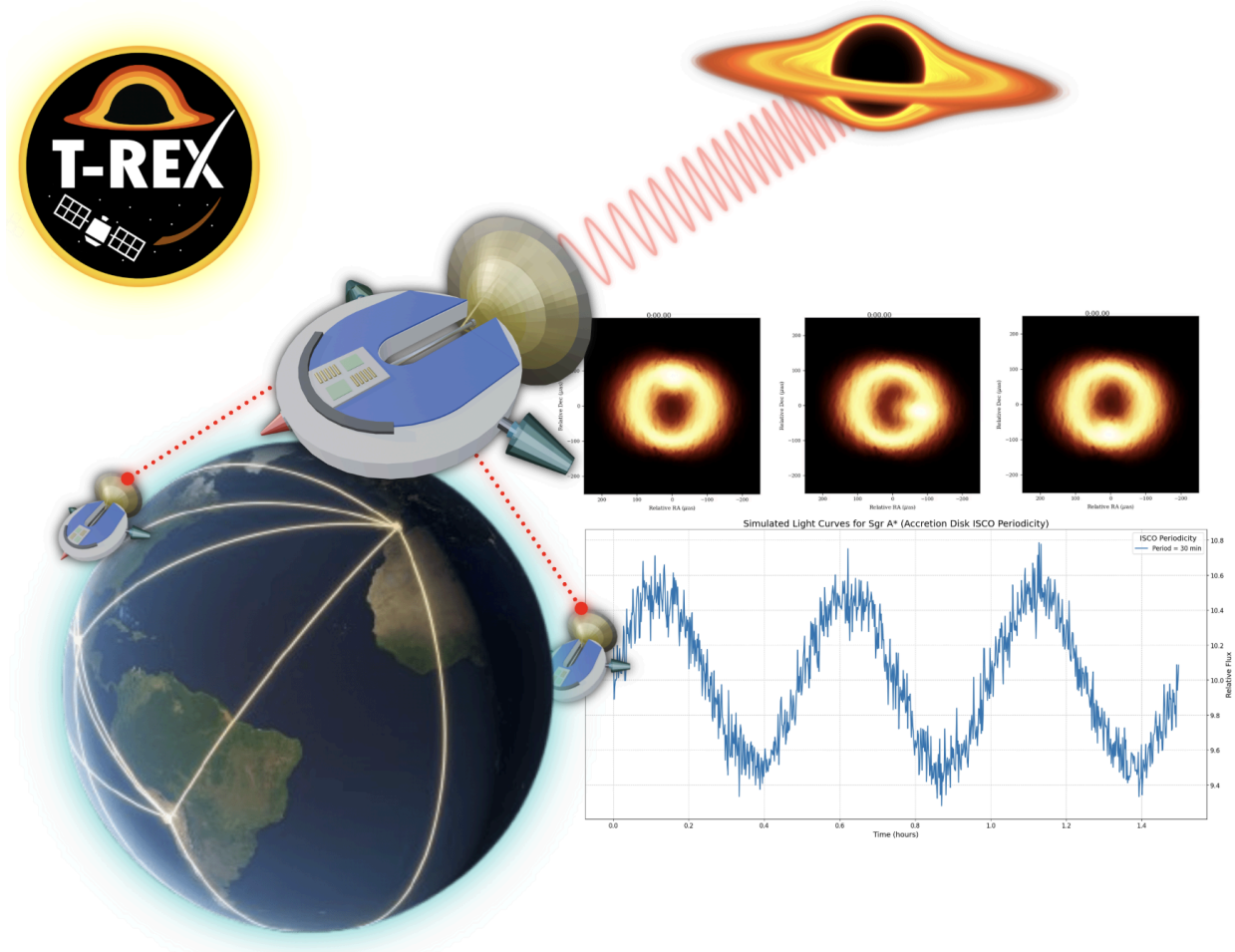


Figure 1: A Closure-Phase Triangle of T-REX DiskSat Interferometers Time-Resolving Sgr A at 86-690 GHz; Measures light curve associated with accretion disk periodicity*

T-REX: Time-Resolving Explorer

To capture the first time-resolved movies of a black hole and binary black hole system will require an instrument with micro-arcsecond angular resolution and sensitivity to flux densities $>40\text{mJy}$. Ground-based interferometry is reaching its sensitivity limits: the EHT has already achieved the largest possible terrestrial baseline of 12,000 km, the South Pole Telescope is establishing a monthly cadence of Sgr A* at 86 GHz, and Next-Generation EHT (ngEHT) is pushing to 345GHz, the maximum Earth's atmosphere will permit. But terrestrial VLBI is limited by its maximum observing frequency, temporal resolution, and field of view. The future of black hole physics demands a VLBI constellation in space. To this end, we propose T-REX: Time-Resolving Explorer, a wide-field, rapid-response, multi-wavelength VLBI constellation in Low-Earth Orbit.

By extending VLBI into the time domain, T-REX will enable the time-resolved observations of Sagittarius A*, M87, active galactic nuclei (AGN), and transient radio phenomena. The spectral energy distribution (SED) of Sgr A* demonstrates that its bolometric luminosity is dominated by radio and millimeter emission, establishing these bands as critical probes of horizon-scale processes. Variability in Sgr A* is now widely recognized as a key diagnostic of dynamics in the immediate vicinity of the black hole. Sgr A* is one of many radio-loud LLAGN/AGNs, for which a time-domain VLBI observatory would enable precision tests of black hole horizon dynamics.

Why ASTRA Now: The present moment is ideal for advancing T-REX into ASTRA:

- 1. Strategic NASA Priorities:** NASA's space telescopes lack a flagship radio observatory. Previous probe-class missions (Space Interferometry Mission & Advanced Radio Interferometry between Space and Earth) were not recommended for further development. The only previous Space VLBI probe-class missions, RadioAstron and VSOP, were led by Russia and Japan, respectively. A NASA-led VLBI constellation would maintain U.S. leadership in black hole astrophysics, establish an ultra-high angular resolution capability observatory and deliver on Astro2020's priorities.
- 2. Technology and Analysis Readiness:** The advancement of NASA's SSDS program has enabled optical downlink from LEO, sub-cm orbital determination via cross-links, and docking technology for nanosatellites have matured to TRL 2.
- 3. Public Engagement:** The first image of a black hole was viewed by 1.4 billion people. The first movie of a black hole and binary black hole system stands to capture public fascination with black holes yet again. T-REX also offers NASA a concrete example of a student-led, mentor-supported mission concept. T-REX was a proposal originally created by students at Brown University, ultimately growing to include collaborators spanning GSFC, Princeton, and JPL.

T-REX: Time-Resolving Explorer

Objectives	Physical Parameters	Observables	Potential Challenges
Probe the physics of supermassive black holes Constrain spin of supermassive black hole candidates (SMBHs) Place constraints on competing accretion models: Magnetically Arrested Disk (MAD) & Standard and Normal Evolution (SANE).	Power spectral analysis of the variable emission will be used to search for periodicities across a broad frequency range in the accretion flow, probing temporal regimes that remain largely unexplored. MAD models, favored by EHT, predict highly magnetized flows. In contrast, SANE models are dominated by short-timescale variability. These differences produce distinct variability signatures.	T-REX will enable long-duration, high-resolution polarimetric monitoring of Sgr A*, providing continuous millimeter observations for 10 hours per day >2 months. This dataset will permit robust characterization of both short- and long-timescale variability. T-REX monitoring will constrain the distribution and duration of variability, providing diagnostics of black hole spin.	ISM scattering, refractive noise, weather interruptions, and sparse ground-station coverage can mimic or obscure intrinsic source variability. Robust cadence, multi-epoch monitoring, and scattering-aware modeling will be required to separate astrophysical variability from propagation and instrumental effects.
Time-resolve binary black hole systems Conduct parameter estimation on supermassive black hole binaries (SMBHBs)	Binary separation, orbital phase/motion, component mass ratio, jet orientation, jet precession, accretion-state variability, and possible long-term orbital evolution.	Time-resolved VLBI imaging of compact binary candidates; mm-light-curve variability; evolving jet morphology; coordinated monitoring of PTA-relevant SMBH binary candidates, including systems of interest to NANOGrav.	SMBHBs evolve over >1 Myr timescales and are thus difficult to spatially resolve, requiring a minimum separation of >2 μas and >40mJy. VLBI nominally has a restrictive <200μas field of view.
Survey of time-domain mm-wavelength targets Determine what powers jets from AGNs	Outflow kinetic energy, shock radius, angular size, expansion velocity, disk–corona–jet coupling timescale, and the nonthermal electron energy distribution.	mm light curves; time-resolved jet morphology; polarization evolution; shock expansion signatures; late-time radio/mm brightening;	AGN jets vary on minute-to-hour timescales — no current facility offers continuous high-cadence mm monitoring

3. Instrument Description

T-REX is a time-domain VLBI observatory. By employing a modular, scalable constellation architecture, it addresses critical NASA Tech Tier Gaps for formation flying and Radio Infrastructure on SmallSats. T-REX consists of DiskSats deployed from a central Pez spacecraft bus. Each DiskSat functions as an independent millimeter-wave radio telescope, carrying a deployable antenna to observe supermassive black holes. Each DiskSat can record VLBI data locally using an 86 GHz receiver chain consisting of a feed horn, orthomode transducer, low-noise amplifiers, frequency conversion electronics, and digital backend. The DiskSats subsequently dock with the Pez, which correlates DiskSat data and RF downlinks to ground stations. The initial architecture will investigate a scalable Pez+DiskSat configuration and validate critical instrumentation.

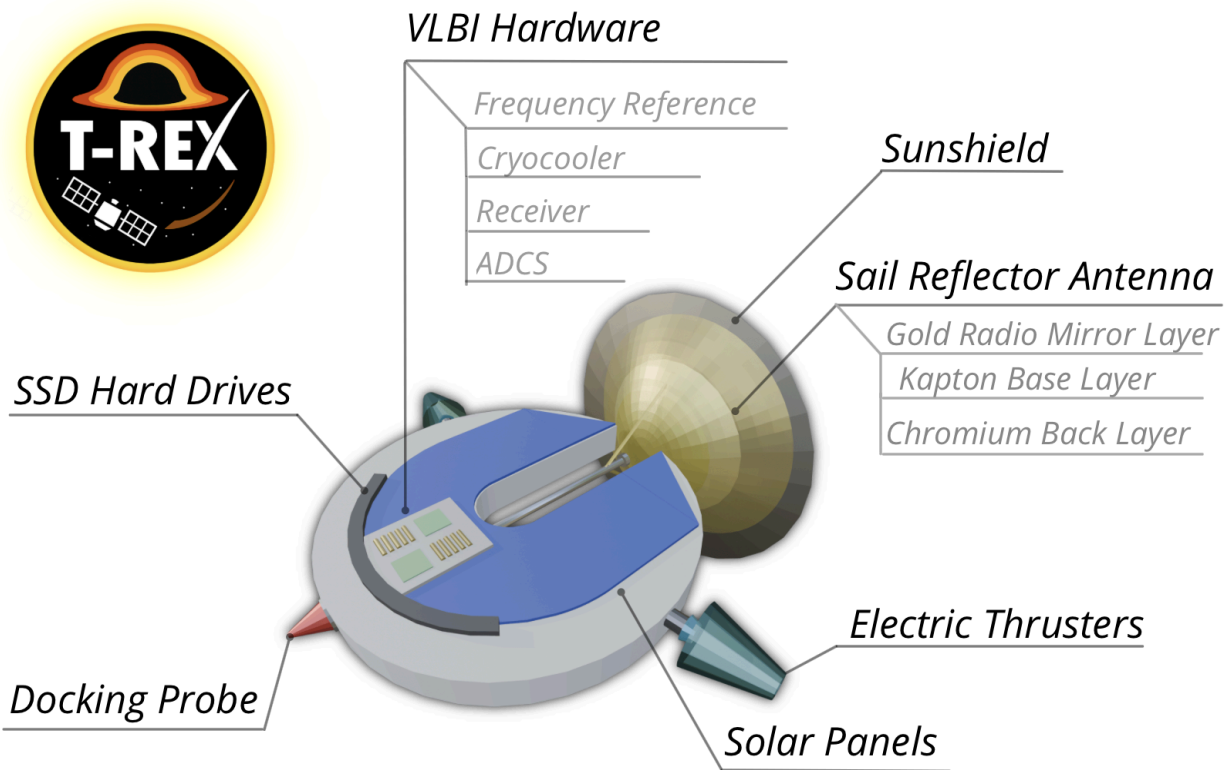


Fig. 2: Instrumentation of Pez DiskSat for 86-690 GHz

The baseline T-REX concept employs a deployable sail-derived member reflector, CRFP reflector or a mesh reflector as the primary structure. Each DiskSat utilizes a lightweight three axis ADCS system, consisting of reaction wheels, magnetorquers, to maintain target pointing and servicing accuracy. Additional technology maturation areas include: autonomous DiskSat deployment and recovery, VLBI correlation, and precision timing synchronization. T-REX would be compatible with existing and future EHT infrastructure, thus complementing existing ground facilities.

4. Mission Implementation

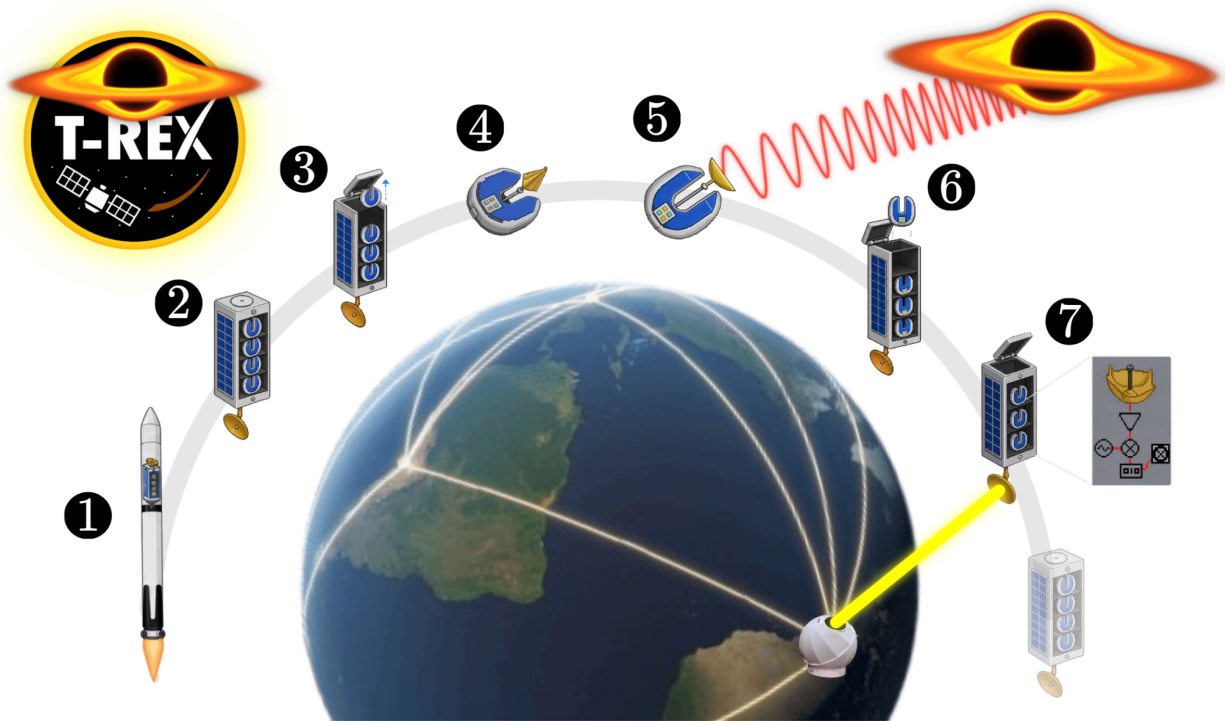


Fig 3: Concept of Operations for T-REX Mission Concept

T-REX consists of two main elements: a “Pez” dispenser and DiskSats deployed from the Pez. T-REX operates with the following nominal concept of operations:

1	Launch	5	DiskSat constellation observes black holes
2	Pez Dispenser in LEO	6	DiskSats rendezvous and docks with Pez
3	Pez deploys DiskSats	7	Pez correlates DiskSat data and downlinks
4	DiskSat antenna deployment	8	Secondary Science

After bus deployment, each DiskSat will independently observe supermassive black holes and relativistic jets using deployable reflector antennas. Observation data will be stored on board and subsequently transferred to the Pez for correlation and downlink. The mission architecture is designed for repeated deployment of both DiskSats and their reflector sails, enabling a reusable distributed VLBI observatory architecture. Key system-level design drivers include precision time synchronization across the distributed VLBI array, high-precision pointing, autonomous Pez-DiskSat docking, and reliable onboard data correlation. T-REX leverages sub-minute temporal resolution, rapidly evolving space-VLBI baselines, and on-orbit correlation to create a rapid-response, time-domain observatory capable of monitoring black holes, relativistic jets, and compact objects as evolving systems at an unprecedented scale.